



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO TMS FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

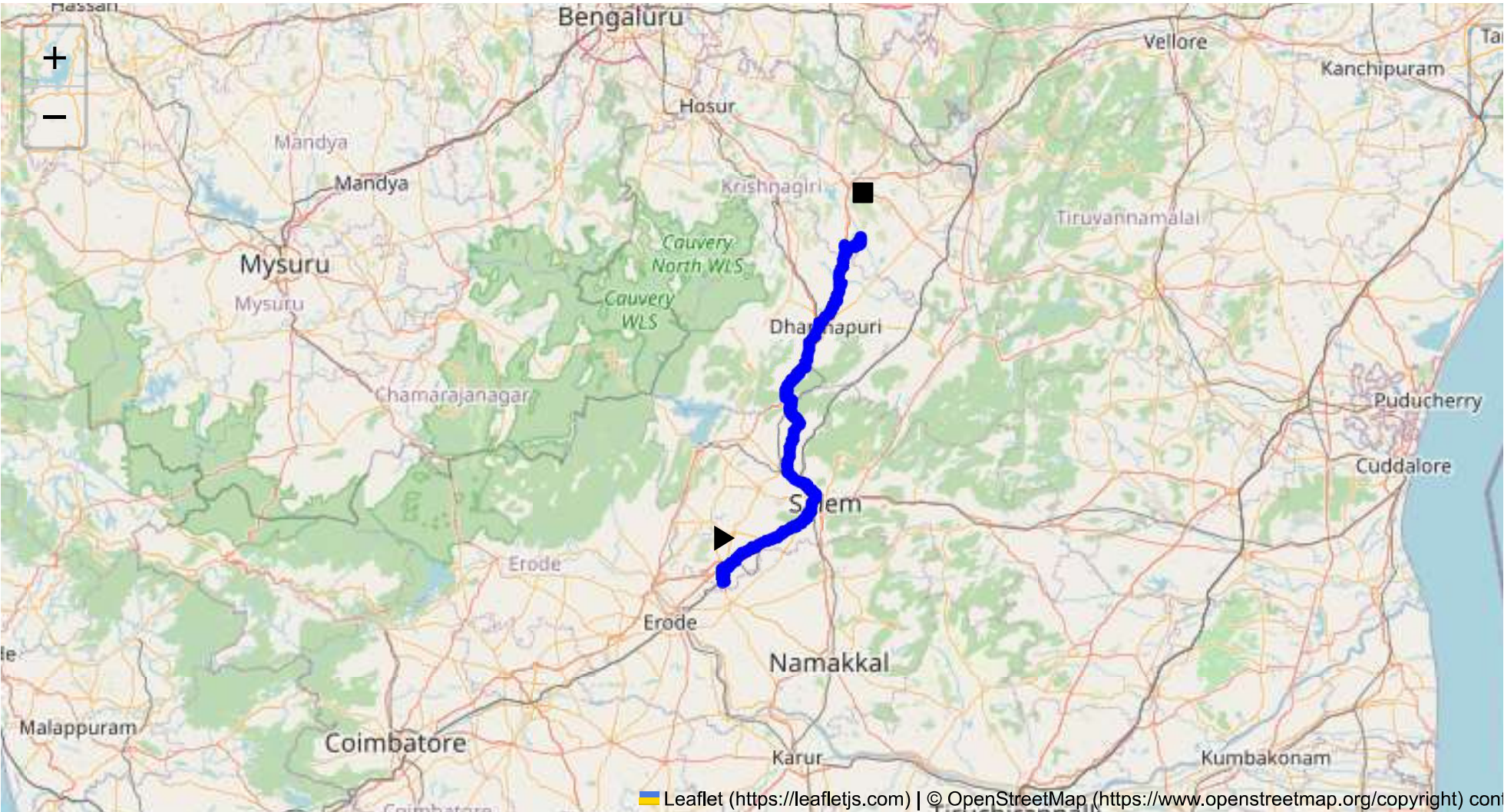
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 140.40 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.3 hours
Start: (11.4381, 77.8734)
End: (12.380099, 78.256557)



Welcome to the Journey Risk Management Study

To analyze the route from CVQF+23W, Sangagiri, Tamil Nadu to 97J4+2J Nedungal, Tamil Nadu, India, I will provide insights into each specified point. Given the nature of this analysis focused on truck drivers carrying hazardous materials, it's essential to ensure thoroughness in each assessment point.

1. Overview of the Route Map:

- The primary route follows major highways and state roads, connecting Sangagiri to Nedungal through a series of interlinked roadways. The route accommodates heavy vehicles and covers a mix of urban and rural landscapes.

2. Typical Weather Conditions and Potential Hazards:

- Tamil Nadu typically experiences hot and humid weather, with significant monsoon rainfall from June to September. During the monsoon, roads can become slippery, and low-lying areas may be prone to flooding, potentially causing delays and accidents.

3. Traffic Patterns and Congestion:

- Peak traffic hours generally occur in the mornings (7:30-9:30 AM) and evenings (5:30-8:00 PM). Congestion is common near urban centers and industrial areas. Planning travel during off-peak hours is advisable to avoid delays.

4. Assessment of Road Quality and Infrastructure:

- The major highways are generally well-maintained, but secondary roads might show varying conditions, including potholes and uneven surfaces. Road signs and lighting can be inconsistent, especially in rural stretches.

5. Suggestions for Alternative Routes for Emergencies:

- Consider NH 44 and SH 20 as potential bypass routes. However, always ensure to verify conditions of these roads ahead of time for size and load restrictions concerning hazardous materials.

6. Local Regulations on Hazardous Material Transport:

- Vehicles carrying hazardous materials must comply with Indian regulations, including possessing proper permits and following designated routes. Keep abreast of any local restrictions or recent changes to compliance requirements.

7. Historical Incidents:

- It's crucial to examine any records of historical incidents involving heavy vehicles or hazardous materials on this route. Possible examples might include spillovers or accidents along narrow or congested areas, although specific instances would need further formal report access.

8. Environmental Considerations and Sensitive Areas:

- The route may intersect environmentally sensitive areas, such as water bodies or agricultural lands. Special attention should be paid to prevent contamination from potential spills, and emergency protocols should be in place.

9. Communication Coverage and Dead Zones:

- While key highways usually have stable phone and data connectivity, rural or mountainous areas may have sporadic coverage. Drivers should be prepared with offline navigation tools and radio communication for continuous contact.

10. Estimated Emergency Response Times:

- Urban areas might have emergency services arrive within 30-45 minutes, while rural or isolated stretches could see delays extending beyond an hour. Ensure knowledge of the nearest emergency facilities along the route.

11. Overall Summary of Risk Assessment:

- The route presents moderate risks typical of highways with potential weather impacts during monsoon seasons and the challenges posed by varying road quality and congestion. Adherence to regulations, cautious driving during peak times, and preparation for communication disruptions are crucial. Planning around weather forecasts and having emergency contingencies can significantly reduce risks.

By considering these aspects, truck drivers can be better prepared and more aware of the potential risks and necessary precautions to safely transport hazardous materials along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	12.3542338, 78.2140141	10 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	High	11.49149, 77.89601	15 KM/Hr
8	Turn	High	11.49224, 77.89601	15 KM/Hr
9	Turn	Medium	11.53501, 77.95707	30 KM/Hr
10	Turn	Medium	11.53601, 77.95856	30 KM/Hr
11	Turn	Medium	11.62403, 78.10893	30 KM/Hr
12	Turn	High	11.64180, 78.11808	15 KM/Hr
13	Turn	High	11.64325, 78.11993	15 KM/Hr
14	Turn	Medium	11.69519, 78.10775	30 KM/Hr
15	Turn	Medium	11.73958, 78.05105	30 KM/Hr
16	Turn	Medium	11.75835, 78.05016	30 KM/Hr
17	Turn	Medium	11.86847, 78.08367	30 KM/Hr
18	Turn	Medium	11.87942, 78.08197	30 KM/Hr
19	Turn	Medium	11.92585, 78.06507	30 KM/Hr
20	Turn	Medium	11.93704, 78.05402	30 KM/Hr
21	Turn	High	11.95818, 78.04978	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
22	Turn	Medium	11.95988, 78.04948	30 KM/Hr
23	Turn	Medium	11.96304, 78.05136	30 KM/Hr
24	Turn	High	11.96399, 78.05135	15 KM/Hr
25	Turn	High	11.96580, 78.04923	15 KM/Hr
26	Turn	Medium	11.96675, 78.04939	30 KM/Hr
27	Turn	Medium	11.96791, 78.05070	30 KM/Hr
28	Turn	Medium	12.19881, 78.18157	30 KM/Hr
29	Turn	Medium	12.22459, 78.19136	30 KM/Hr
30	Turn	Medium	12.25060, 78.19977	30 KM/Hr
31	Turn	Medium	12.25329, 78.20039	30 KM/Hr
32	Turn	Medium	12.29110, 78.19909	30 KM/Hr
33	Blind Spot	Blind Spot	12.35421, 78.21413	10 KM/Hr
34	Turn	High	12.35098, 78.21508	15 KM/Hr
35	Turn	High	12.35143, 78.21943	15 KM/Hr
36	Turn	High	12.35212, 78.21983	15 KM/Hr
37	Turn	Medium	12.35240, 78.22112	30 KM/Hr
38	Turn	Medium	12.35096, 78.22562	30 KM/Hr
39	Turn	Medium	12.35028, 78.22631	30 KM/Hr
40	Turn	Medium	12.34951, 78.23507	30 KM/Hr
41	Turn	Medium	12.35287, 78.24038	30 KM/Hr
42	Blind Spot	Blind Spot	12.35668, 78.24491	10 KM/Hr
43	Turn	Medium	12.35541, 78.24636	30 KM/Hr
44	Blind Spot	Blind Spot	12.35537, 78.24726	10 KM/Hr
45	Turn	High	12.35617, 78.24719	15 KM/Hr
46	Turn	High	12.35652, 78.24749	15 KM/Hr
47	Turn	Medium	12.35575, 78.24943	30 KM/Hr
48	Turn	High	12.35624, 78.25185	15 KM/Hr
49	Turn	High	12.36447, 78.25342	15 KM/Hr
50	Turn	High	12.36512, 78.25221	15 KM/Hr
51	Turn	High	12.36649, 78.25297	15 KM/Hr
52	Turn	Medium	12.36738, 78.25260	30 KM/Hr
53	Turn	High	12.37241, 78.25456	15 KM/Hr
54	Turn	High	12.37420, 78.25743	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
20	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
21	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
22	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
24	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
25	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
26	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
27	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
28	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
31	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
32	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
33	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
34	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
35	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
37	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
23	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
29	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
30	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
36	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
38	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 12.3542338, 78.2140141



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64325, 78.11993



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69519, 78.10775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73958, 78.05105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93704, 78.05402



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96580, 78.04923



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22459, 78.19136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25060, 78.19977



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25329, 78.20039



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29110, 78.19909



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.35421, 78.21413



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.35098, 78.21508



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.35143, 78.21943



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.35212, 78.21983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.35240, 78.22112



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.35096, 78.22562



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.35028, 78.22631



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.34951, 78.23507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.35287, 78.24038



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.35668, 78.24491



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.35541, 78.24636



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.35537, 78.24726



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.35617, 78.24719



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.35652, 78.24749



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.35575, 78.24943



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.35624, 78.25185



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.36447, 78.25342



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.36512, 78.25221



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.36649, 78.25297



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.36738, 78.25260



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.37241, 78.25456



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.37420, 78.25743

